MULTIMEDIA:

→ The technology in which multiple content types (Audio, Video, Text, image, Animation) are combined and presented.

→ Integration of different forms of content types.

→ multimedia is classified into 4 categories.
(a) Text (b) Images (c) Audio (d) video.

* Applications:

→ Video: Filmora, VN, Final Cut pro ..

→ Audio: Audacity, Adobe Audition.

→ Image: Canva, Photoshop

→ Animation:- Adobe Animate, Blender.

→ Interactive app:- Web apps.

* multimedia hardware:-

→ Devices that are used to store, process and display multimedia content.

eg:- Graphic card, Sound card, motherboard.

* multimedia Audio:-

1) Digital medium:- Storing sound audio in digital form.
→ Audio is used in the form of continuous waves.

2) Digital Audio tech:- Sound wave converted into binary form (0-1)

Two terms:-

(a) Sampling :- Converting audio signals into discrete value.

(b) Bit Depth :- Improving and defining the quality of the audio signal.

* Classification of multimedia :-

(a) Linear :-
→ A user has no control over it, it can only be watched or listened to. eg: movie, animated video.

(b) Non-linear :-
→ The user can interact with it.
eg: E-learning app, video game.

* Text & Graphic in multimedia :-

* Text in multimedia
→ Text.
Types :-
(a) Static → Fixed text, eg: Heading & Paragraph
(b) Dynamic → Scrolling text, Animated text.

* Graphics in multimedia :

(a) Color :
→ Primary colors → (RGB)

Colour model :
(i) RGB :- for display
(ii) CMYK (Cyan, Magenta, Yellow, Black). for printing.

(b) Digital Image :-

→ Image = Grid of Pixels.
→ Each Pixel has a colour.
→ Pixel = Smallest unit of an image (dot)
→ The Quantity of any display device (monitor) is measured by its resolution.
→ Resolution = Total no. of pixel in screen.
→ Unit of Resolution = DPI (Dot per inch) / PPI (Pixel per inch)
→ The ratio of horizontal and vertical pixel on any screen = Aspect Ratio :- 16:9.
→ The distance b/w two pixel = Dot pitch.
→ Resolution ↑, dot pitch ↓
→ Two factors affect Image Quality.
(a) Resolution
(b) Color depth.

*

Feature ↓	lossless	lossy
meaning	Compressed without Quality loss	Some of the data is lost permanently
Quality	same	Reduced
file size	moderate to same	Reduced
Use case	Document, medical image.	web image, music, video.
Example	PDF, CDR, PNG, ZIP, Tiff	JPG, MP3, MP4

* Quality important → lossless
* Storage important → lossy.

* Raster v/s vector:-

→ Raster images are made up of dots/pixels, whereas vector images are HD graphics.
→ Raster images are called bitmap pictures.

* Sound in multimedia:-

→ Sound cards are used to process audio signals.
→ It works as an interface b/w input device (mic) and output device (speaker)
→ The sound card also used ADC (Analog to Digital converter) or DAC (Digital to Analog)
→ The mic converts sound wave to electrical signal.
→ Sampling rate defines how many samples of sound are captured in one second.
eg 44.1 kHz (CD), 48 kHz (TV)
→ Bit depth defines audio quality.
more bit depth = more detailed audio.

* MPEG :-

→ moving Picture Experts Group.
→ Standard setting group that creates formats for digital video & audio.
→ lossy Compression reduce Quality a little bit greatly reduces file size.
→ MPEG formats store video, audio, and subtitle in a single file.

(a) MPEG-1 :-

→ Extention : .mpg.
→ Supports video upto 30fps (frames per second)
→ Codec (Coder-Decoder) : MP3 (MPEG-1 Audio layer-3)
→ frames per second : Number of images made in one second.

(b) MPEG-2 :-

→ Higher Quality than MPEG-1.
→ Used in DVDs, Cable TV.
→ Formats : .m2v, .mpg, .VOB (video objects).

(c) MPEG-4 :-

→ Higher Quality than MPEG-2
→ Supports Audio + video + text + 3D objects.
→ Popular format : .mp4
→ Used in OTT apps, Youtube, mobile videos.

Lower the file.

formats:-

→ .mpg / .mpeg ⇒ Used in MPEG-1 & MPEG-2 video.

→ .mp4 ⇒ MPEG-4 Audio + video

→ .mp3 ⇒ MPEG-1 audio layer-3.

→ .m4v ⇒ .mp4 version for Apple devices.

→ .m2v ⇒ MPEG-2 video (no audio).

→ .ts ⇒ Transport Stream (Broadcasting), MPEG-2.

* MIDI:-

→ Musical Instrument Digital Interface.

→ MIDI is a protocol that connects musical instruments to Computers.

→ MIDI does not record sound; it only transmits pitch, timing, and intensity as digital signals.

* Audio formats:-

- WAV = Waveform Audio file format.
- AIFF = Audio Interchange file format.
- FLAC = free lossless Audio Codec.
- AAC = Advanced Audio Codec.